

IPs asignadas a los Routers

Conexión R-CORE a R-N2

Red: 172.20.0.144/30

Dispositivo	IP
R-CORE	172.20.0.145
R-N2	172.20.0.146

Conexión R-CORE a R-N3

Red: 172.20.0.148/30

Dispositivo	IP
R-CORE	172.20.0.149
R-N3	172.20.0.150

Conexión R-N2 a R-N3

Red: 172.20.0.152/30

Dispositivo	IP
R-N2	172.20.0.153
R-N3	172.20.0.154

Conexión entre routers

R-CORE

```
configure terminal
```

```
interface fastEthernet 1/0
```

```
ip address 172.20.0.145 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
interface serial 2/0
```

```
ip address 172.20.0.149 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
end
```

```
write memory
```

R-N2

```
configure terminal
```

```
interface fastEthernet 0/0
```

```
ip address 172.20.0.146 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
interface fastEthernet 1/0
```

```
ip address 172.20.0.153 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
end
```

```
write memory
```

R-N3

```
configure terminal
```

```
interface serial 2/0
```

```
ip address 172.20.0.146 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
interface fastEthernet 1/0
```

```
ip address 172.20.0.154 255.255.255.252
```

```
no shutdown
```

```
exit
```

```
end
```

```
write memory
```

Switches

SW-N1

VLAN	Nombre	Puerto
10	VEN	2
10	VEN	3
10	VEN	4
10	VEN	5
10	VEN	6
20	REC	7
20	REC	8
20	REC	9
20	REC	10
20	REC	11
30	ACL	12

VLAN	Nombre	Puerto
30	ACL	13
30	ACL	14
30	ACL	15
30	ACL	16
40	ESP	17

Configurando R-CORE para trabajar con los VLANs de SW-N1

```
configure terminal
```

```
interface FastEthernet0/1  
no ip address  
no shutdown  
exit
```

```
interface FastEthernet0/1.10  
encapsulation dot1Q 10  
ip address 172.20.0.73 255.255.255.248  
exit
```

```
interface FastEthernet0/1.20  
encapsulation dot1Q 20  
ip address 172.20.0.97 255.255.255.248  
exit
```

```
interface FastEthernet0/1.30  
encapsulation dot1Q 30  
ip address 172.20.0.89 255.255.255.248  
exit
```

```
interface FastEthernet0/1.40  
encapsulation dot1Q 40
```

```
ip address 172.20.0.1 255.255.255.240
exit

end
write memory
```

SW-N2

VLAN	Nombre	Puerto
50	ADM	2
50	ADM	3
50	ADM	4
50	ADM	5
50	ADM	6
60	CON	7
60	CON	8
60	CON	9
60	CON	10
60	CON	11
70	RHH	12
70	RHH	13
70	RHH	14
70	RHH	15
70	RHH	16
80	SRN	17
80	SRN	18
90	CAM	19
90	CAM	20

VLAN	Nombre	Puerto
90	CAM	21
90	CAM	22
90	CAM	23

Configurando R-N2 para trabajar con los VLANs de SW-N2

```
configure terminal
```

```
interface FastEthernet0/1  
no ip address  
no shutdown  
exit
```

```
interface FastEthernet0/1.50  
encapsulation dot1Q 50  
ip address 172.20.0.81 255.255.255.248  
exit
```

```
interface FastEthernet0/1.60  
encapsulation dot1Q 60  
ip address 172.20.0.105 255.255.255.248  
exit
```

```
interface FastEthernet0/1.70  
encapsulation dot1Q 70  
ip address 172.20.0.113 255.255.255.248  
exit
```

```
interface FastEthernet0/1.80  
encapsulation dot1Q 80  
ip address 172.20.0.33 255.255.255.240  
exit
```

```
interface FastEthernet0/1.90
encapsulation dot1Q 90
ip address 172.20.0.65 255.255.255.248
exit

end
write memory
```

SW-N3-01

VLAN	Nombre	Puerto
100	GER	1
100	GER	2
100	GER	3
100	GER	4
100	GER	5
110	IT	6
110	IT	7
110	IT	8
110	IT	9
110	IT	10
120	SOP	11
120	SOP	12
120	SOP	13
120	SOP	14
120	SOP	15
130	SJU	16
130	SJU	17

Configurando R-N3-01 para trabajar con los VLANs de SW-N3

```
configure terminal

interface FastEthernet0/1
no ip address
no shutdown
exit

interface FastEthernet0/1.100
encapsulation dot1Q 100
ip address 172.20.0.121 255.255.255.248
exit

interface FastEthernet0/1.110
encapsulation dot1Q 110
ip address 172.20.0.129 255.255.255.248
exit

interface FastEthernet0/1.120
encapsulation dot1Q 120
ip address 172.20.0.137 255.255.255.248
exit

interface FastEthernet0/1.130
encapsulation dot1Q 130
ip address 172.20.0.17 255.255.255.240
exit

end
write memory
```

Configurando HSPR para que los servidores estén conectados a los 3 routers

R-CORE

```
configure terminal

interface FastEthernet0/1.140
encapsulation dot1Q 140
ip address 172.20.0.50 255.255.255.240
standby 1 ip 172.20.0.49
standby 1 priority 110
standby 1 preempt
exit

end
write memory
```

R-N2

```
configure terminal

interface FastEthernet0/1.140
encapsulation dot1Q 140
ip address 172.20.0.51 255.255.255.240
standby 1 ip 172.20.0.49
standby 1 priority 100
exit

end
write memory
```

R-N3

```
configure terminal

interface FastEthernet0/1.140
encapsulation dot1Q 140
```

```
ip address 172.20.0.52 255.255.255.240
standby 1 ip 172.20.0.49
standby 1 priority 90
exit

end
write memory
```

Enrutamiento Dinámico con OSPF

R-CORE

```
configure terminal

router ospf 1

network 172.20.0.72 0.0.0.7 area 0
network 172.20.0.96 0.0.0.7 area 0
network 172.20.0.88 0.0.0.7 area 0
network 172.20.0.0 0.0.0.15 area 0

network 172.20.0.48 0.0.0.7 area 0

network 172.20.0.144 0.0.0.3 area 0
network 172.20.0.148 0.0.0.3 area 0
exit

end
write memory
```

R-N2

```
configure terminal

router ospf 1
```

```
network 172.20.0.80 0.0.0.7 area 0
network 172.20.0.104 0.0.0.7 area 0
network 172.20.0.112 0.0.0.7 area 0
network 172.20.0.32 0.0.0.15 area 0
network 172.20.0.64 0.0.0.7 area 0

network 172.20.0.48 0.0.0.7 area 0

network 172.20.0.144 0.0.0.3 area 0
network 172.20.0.152 0.0.0.3 area 0
exit

end
write memory
```

R-N3

```
configure terminal

router opsf 1

network 172.20.0.120 0.0.0.7 area 0
network 172.20.0.128 0.0.0.7 area 0
network 172.20.0.136 0.0.0.7 area 0
network 172.20.0.16 0.0.0.15 area 0

network 172.20.0.48 0.0.0.7 area 0

network 172.20.0.148 0.0.0.3 area 0
network 172.20.0.152 0.0.0.3 area 0
exit

end
write memory
```